

followup

Pay with your thumb!

A practical application of the Sense and Sensibility story carried in the July 2002 issue is seen in Texas where three Kroger stores are testing biometric systems that enable shoppers to pay with their thumbprint. Customers register with a store by placing their thumb on an electronic screen that records the print. The information is encrypted and linked to the customer's credit card, debit card or checking account information. A picture ID establishes the shop-



per's identity and the whole process takes just 20 seconds! Though more technology-savvy people were expected to be most interested, everyone from teenagers to the elderly are sign-

ing up. Grocers say that controlling fraud was their main goal—credit and debit card fraud cost grocery chains an average of \$1,96,000 each in 2000. Though the feedback is positive, reports show a fear of government and law surveillance. Kroger employees remind shoppers that the system is based on the thumbprint rather than the index finger that is standard in law enforcement. Also, they don't store the entire thumbprint, just crucial characteristics.

Microsoft to reveal Windows code

Microsoft Corporation will reveal hundreds of pieces of proprietary code from its Windows operating system to comply with an antitrust settlement it signed in November 2001 with the US Justice Department. It will disclose 385 modules of internal operating rules, previously kept secret. The Justice Department and 18 states had accused Microsoft of deliberately withholding computer code in Windows to hamper competitors. Microsoft reached a settlement and nine states agreed to sign, but nine others are still asking US District Judge Colleen Kollar-Kotelly for tougher restrictions.

snapshot

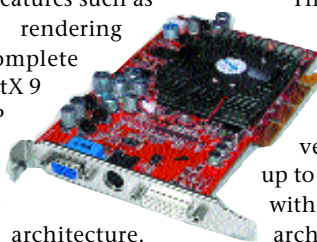
Dataquest predicts
1 GB memory for
entry-level PCs by
2005

Source: Dataquest

Graphics at the speed of light

ATI Technologies Inc has introduced the RADEON 9700 Visual Processing Unit (VPU), a groundbreaking chip that boasts of features such as eight parallel rendering pipelines, complete Microsoft DirectX 9 support, AGP 8x support and a fully programmable floating-point architecture. The chip's eight parallel rendering pipelines provide the brute force to process over 2.5 billion pixels per second. The four parallel geometry engines process greater than 300 million transformed and lit polygons per second. In addition, Hyper Z III reduces overdraw by detecting and discarding hidden pixels and reduces memory bandwidth consumption by over 50 per cent. It incorporates a new high-performance 256-bit DDR memory interface, providing over 20 Gbps of graphics memory bandwidth and fully supports the AGP 8x specification, allowing increased volumes of texture and vertex data to be transferred quickly.

The AGP 8x interface was designed to double the graphics processing bandwidth over today's AGP 4x.



The chip supports up to 16 textures per rendering pass with Pixel Shaders 2.0 and vertex programs of up to 1,024 instructions with flow control. This architecture enables users to run complex cinematic-quality shader programs in real time for truly realistic effects. SmoothVision 2.0 incorporates a full-scene anti-aliasing implementation for unprecedented performance at even the highest quality anti-aliasing modes. Also included is ATI's popular advanced anisotropic filtering technology, supporting up to 16 bilinear or trilinear samples per pixel in a variety of modes. These technologies substantially improve the visual quality, without degrading performance. ATI's retail product will ship with 128 MB DDR memory, at a suggested retail price of \$399 (approx Rs 20,000).



heroes

Open Source

nVidia will make its Cg development tools for programming high-level graphics effects open-source software. RealNetworks too plans to publicly release the code underlying its multimedia player software, though certain details may change before its release.

Adult entertainment

The US adult Web site industry has set up a trade organisation to represent it. The Internet Freedom Association (IFA) aims to promote the rights of the online adult entertainment industry, while taking a strong stand against child pornography.



zeroes

Privacy

Japan launched a compulsory government ID system in the face of stiff protests calling it a violation of privacy. The database stores names, addresses, dates of birth, gender and the new ID numbers for each of Japan's 127 million citizens, making it easier for them to obtain documents for a variety of public services and benefits.

Piracy

Federal prosecutors charged Canadian Robert Garby with selling of illegal mod chips and pirated console games through his Kustom Komputers business. He pleaded guilty to six counts of criminal code violations and copyright law infringement and was fined \$17,000 and sentenced to a year of probation.